Foundations: Agents and Automation, Explained

Instructor guide · Day 1 · Session 1

What agents and automations are, the four parts of an agent, workflow versus agent, real admin examples

Session at a glance

Day / session	Day 1 · Session 1 · 120 minutes
Lead instructors	Mousa AL-Akhras, Mohammed Abdelmajeed
Online module	https://albanna-tutorials.com/ai-administrator/01_agents_and_automation.html
Deck	Day1_S1_Foundations.pptx, 72 slides, speaker notes included

Learning objectives

- Explain the difference between an automation (fixed steps) and an agent (a model that decides its own steps).
- Name the four building blocks of an agent: model, tools, memory and instructions.
- Describe the think, act, observe loop and why it needs an iteration limit and good tools.
- Recognise the three designs used in this course: automation, workflow with AI steps, agent.
- List three tasks from their own job that suit automation, and one that should stay with a person.

Run of show

Minutes	Block	Focus
3 to 10	1. Why administrators are learning this now	The boundary of what software can do just moved
10 to 35	2. Automation versus agent	Who decides the steps?
35 to 60	3. The four parts of every agent	Model · Tools · Memory · Instructions
70 to 85	4. How an agent actually works: the loop	Think · Act · Observe · Repeat
85 to 90	5. Examples from administration and operations	Which design fits which task, and what the human still does

90 to 108	Practical: Meet a real agent	Talk to a live always-on agent from your phone. Nothing to install.
108 to 116	Practical lab: your task inventory	Steps shown now, finish at home, due before the next session
116 to 120	Knowledge check	Five questions in the style of the final exam

The welcome, objectives and run-of-show slides take the first minutes before Section 1. The break is planned at the midpoint (see the run-of-show slide in the deck). Cuts if running behind are listed under Troubleshooting.

Materials checklist

- Projector or screen share; this deck (Day1_S1_Foundations.pptx) with speaker notes open.
- One AI assistant account logged in and tested on the teaching machine (ChatGPT, Claude, Copilot or Gemini). Ideally two tabs with two different assistants for the demo.
- Practical worksheet (Day1_S1_Practical_Meet_an_Agent.pdf) shared in the chat at minute 90; lab handout (Day1_S1_Lab_Handout.pdf) at minute 108.
- Hermes Agent gateway started on the course server with the Telegram bot live (Day1_S1_Hermes_Setup_Guide.pdf); phone with Telegram for a test message; the bot QR on the Join slide.
- Fallback: the two-minute screen recording of the four tasks, in case Wi-Fi, Telegram or the model provider fails.
- A blank spreadsheet template with the four lab columns (Task, Frequency, Minutes, Judgement) ready to share.
- Quiz bank (Day1_S1_Quiz_Bank.pdf) for the instructor only; five questions are also on slides.
- Participants: an AI assistant account and a blank document or spreadsheet. Nothing to install.

Before the session

- Read the live module once (link above); the slides follow its five sections in order.
- The day before: follow Day1_S1_Hermes_Setup_Guide.pdf (install Hermes, free Gemini key, persona in SOUL.md, web search only, Telegram bot, test the four tasks yourself, record the fallback video).
- Paste your bot QR image as brand/bot_qr.png and rebuild, or paste it onto the Join slide by hand; put the real bot username on that slide.
- Test that an assistant can take a pasted table (the lab's step 4 prompt).
- Decide the break time with the co-instructor (planned at minute 60).
- Ask participants in advance to bring a list of what they do in a typical week.

Practical: Meet a real agent (minutes 90 to 108)

1. Minute 88: hermes gateway status; start it if needed. Send one test message from your phone.
2. Minute 90: 'Join the agent' slide. Participants scan the QR, press Start, say hello. Help anyone without Telegram pair with a neighbour.

3. Minute 93: 'Four tasks' slide and the worksheet PDF in the chat. Let them work. Watch the gateway terminal; read one search tool call aloud.
4. Minute 103: 'What you saw' debrief. One observation per task, mapped to model, tools, memory, instructions and the loop. If Task 4 broke the guard for anyone, celebrate it and point to Day 6.
5. Minute 108: callout slide, then straight into the lab kickoff. After the session: hermes gateway stop and delete ~/.hermes/memories/USER.md; the server stays up for the rest of the course.
6. Optional filler if the practical cannot run: the same admin email in two assistants (ChatGPT and Claude) with 'Summarise this and tell me what to do', then with four labelled output lines. Point out the shape changes and that neither can act: the missing part is tools.

Slide-by-slide talking points

Slide numbers match the deck. Quiz questions occupy two slides each (question, then answer).

1. Agents and Automation, Explained · **Title**
Welcome participants. Introduce yourself and the co-instructor. State the promise of the session in one sentence: by the end of two hours everyone can tell an automation from an agent, name the four parts of any agent, and will have a list of their own tasks to automate during the course. No code today, and no tools to install.

2. Where this session sits · **Bullets**
Give the shape of the whole course so today's vocabulary has somewhere to land. Mention that the final exam is 40 multiple-choice questions plus a timed practical in n8n, and that the knowledge checks at the end of every session are written in the same style.

3. What you will be able to do after this session · **Bullets**
Read the objectives quickly. The last one is the lab deliverable: a one-page task inventory. Everything else in the course builds on that inventory, so it matters more than the quiz.

4. Run of show (120 minutes) · **Table**
Show this for ten seconds. Participants like knowing when the break is. The lab is shown in the room and finished at home, which is fine. The deliverable is due before Day 2.

5. 1. Why administrators are learning this now · **Section** · **minute 3**

6. Office work is a chain of small decisions · **Bullets**
Make it personal: ask two participants to name the first three things they did at work this morning. Almost always it is reading something, deciding something, and copying something between systems. Those three verbs are exactly what language models added to the automatable set.

7. What a language model actually does: predict the next word · **Figure** · **figure: llm-token-prediction**
Keep this to one minute. Participants do not need to know how the model is trained. They need to know that it produces plausible continuations, because that single fact explains both its strengths (fluent reading and writing) and its main weakness (confident, plausible, wrong answers), which we meet in Session 2.

8. A conversation is the same trick, repeated · **Figure** · **figure: llm-conversation-example**
Point out that the model does not 'remember' in the human sense; the entire conversation is handed back to it every turn. That is why long chats slow down and why, later, we will talk about memory as something you design rather than something the model has.

9. Alone, the model forgets · Figure · figure: llm-memory-limitation
This is the first of the two limitations that make the four-part agent necessary. Ask: 'How would you brief a temp who forgets everything overnight?' You would write it down. That written briefing is the memory and instructions we design.

10. Alone, the model guesses at facts and numbers · Figure · figure: llm-math-limitation
Second limitation, second part of the agent. Tools are how the model stops guessing and starts looking things up. For administrators the message is: never ask a bare model for a number that exists in a system.

11. The job of an AI administrator: four verbs · Bullets
This slide is the course in one breath. Repeat the four verbs out loud: map, design, connect, control. Stress that the shift of power is real: the process owner, not the programmer, now decides.

12. 2. Automation versus agent · Section · minute 10

13. Two different things · Two columns
The single question to ask about any system is 'who decides the steps?'. If a person wrote them down, it is an automation. If the model chooses at run time, it is an agent. Everything else in the comparison table follows from that one difference.

14. Automation versus agent, side by side · Table
Walk the rows. Dwell on 'failure style': an automation that breaks stops, an agent that is confused keeps acting. That is why agents need the controls we design on Day 5. Mention the tool names so they are familiar when they appear on Day 3 and Day 4, but do not demo them today.

15. Automation in a picture: a fixed order of tools · Figure · figure: fixed-tool-order
Connect the picture to the definition: the arrows were drawn by a person. Nothing the model does can change the order. This is what you build in n8n on Day 3.

16. Agent in a picture: the model chooses the tool · Figure · figure: autonomous-tool-selection
Same tools, different wiring: now the model is in charge of the arrows. Ask the room which picture they would trust with sending an email to a customer. Most say the first; that intuition is correct and is the point of the rule of thumb coming up.

17. The middle ground: workflows with AI steps · Bullets
This is the most important design idea of the day. Most 'AI automation' that works in real offices is exactly this: n8n or Power Automate with one AI node. Participants should leave able to say 'workflow with an AI step' as a third category, not just automation or agent.

18. Autonomy is a dial, not a switch · Figure · figure: agent-autonomy-levels
Use the levels to place the three designs: automation sits at the fixed end, workflow with AI step in the middle, agent at the autonomous end. Where the human sits on this dial is Day 5's whole topic (controls), so just plant the idea now.

19. Rule of thumb · Callout
Have participants write this down. It is the answer to roughly a quarter of the exam questions, and it is what separates an administrator who ships reliable automations from one who demos impressive agents that break.

20. 3. The four parts of every agent · Section · minute 35

21. An agent, in the most general picture · Figure · figure: agent-framework
Classic agent picture from AI textbooks: sensors in, actuators out, a brain in the middle. For us the 'environment' is email, spreadsheets, ticketing systems, and the brain is the language model. Whatever product you buy, you are buying this picture.

22. The augmented language model · Figure · figure: augmented-llm
This slide fixes the two limitations shown earlier. Forgetting is fixed by memory and retrieval; guessing is fixed by tools. 'Augmented LLM' is the term used in Anthropic's 'Building Effective Agents' guide, which is in the references.

23. The four parts wired together · Figure · figure: agent-components
Spend time here. Ask participants to find each of the four parts in the picture. Learn to see them and every vendor demo becomes readable: 'which model, which tools, what memory, and who wrote the instructions?'

24. Part 1: the model · Figure · figure: augmented-llm-framework
Administrators rarely train or tune models. They choose one. The choice criteria are the bullets. Data residency often decides it in regulated organisations; park that for Day 6 but name it now.

25. Part 2: tools, by type · Figure · figure: tool-types
Give admin examples for each type: search the web, read a calendar, look up a customer, send an email, create a ticket. Tools are also where the risk is: a read-only tool cannot do damage; a 'send email' tool can.

26. A tool call, step by step · Figure · figure: tool-use-example
Narrate the figure left to right. The important moment is the hand-back: the tool result enters the conversation, and the model continues as if it had always known. This is also why tool quality is answer quality: a stale price list produces a confident wrong quote.

27. How the model asks for a tool: a structured request · Figure · figure: json-tool-call
You are not teaching JSON. The point is that the model does not 'do' anything itself; it writes a request, and the platform (n8n, Copilot Studio) executes it. That boundary is where approval gates live on Day 5: the request can be shown to a human before it runs.

28. The platform handles the plumbing · Figure · figure: function-calling
One sentence: 'function calling' is the vendor term for the previous slide. When you see it in a product menu, you now know what it means. Move on.

29. Models learned to use tools (Toolformer, 2023) · Figures
Optional history slide; thirty seconds. It explains why every modern assistant can use tools: it was taught to recognise the moment a tool would help. Skip if behind schedule.

30. Preview: one plug for every tool (Day 4) · Figure · figure: mcp-components
Plant the idea only. On Day 4 participants will connect tools through connectors and MCP servers without code. Today the message is: tools are becoming plug-and-play, so the administrator's job is choosing and permitting them, not building them.

31. Part 3: memory, by type · Figure · figure: memory-types
Memory is what makes an assistant feel like it knows your organisation. The types matter because they are priced and controlled differently: conversation memory is free and transient; a document knowledge base is a design decision with privacy consequences.

32. Short-term memory is the context window · Figure · figure: context-window
Return to the conversation slide: the context window is the whole history handed back each turn. It has a size limit, so long documents and long conversations must be summarised or retrieved in pieces. Session 2 uses this to explain why you give the model the source rather than hope it knows.

33. Long-term memory is stored outside and fetched when needed · Figure · figure: long-term-memory
This is the mechanism behind 'Projects' and 'custom GPTs' with uploaded files, which participants build in Session 2, and behind RAG on Day 4. Keep it conceptual: a filing cabinet the model can search.

34. Part 4: instructions, your main control surface · Bullets
Of the four parts, this is the one administrators own. The whole of Session 2 is about writing these well. Ask: 'If a new temp started tomorrow, what would your one-page briefing say?' That briefing is the system prompt.

35. Always-on personal agents add two ideas · Bullets
Participants may have seen these on social media. Acknowledge them, place them in the framework, and promise the demo on Day 5. Do not install anything today.

36. 4. How an agent actually works: the loop · Section · minute 70

37. The agent loop · Steps
Act it out with the co-instructor: one plays the model, one plays the tool. Goal: 'Find why invoice 4471 was rejected.' Model: 'Look up invoice 4471.' Tool: 'Rejected: PO number missing.' Model: 'Draft a corrected invoice with PO field.' Three loops, done. Then ask: what if the tool had said 'not found'?

38. ReAct: reason, then act · Figure · figure: react-framework
Name the pattern: ReAct, from the 2023 paper in the references. Participants will see 'ReAct agent' as an option in n8n's AI Agent node on Day 4. They only need to recognise the name and know it means this loop.

39. The cycle, as the platform runs it · Figure · figure: react-cycle
Emphasise the stop condition. The loop has no natural end; something must say 'enough'. Either the model decides it has the answer, or a limit set by the administrator fires. Lead into the next slide.

40. Planning is deciding the next step, every step · Figure · figure: llm-planning-behavior
Planning sounds grand; in practice it is this loop plus the model's ability to break a goal into sub-steps. Good instructions make planning better: 'Always check the policy document before answering' is a planning instruction.

41. Two consequences for administrators · Two columns
These two are exam material and real-world material. The iteration limit is the first control anyone sets on an agent. The tool-quality point saves weeks of prompt fiddling: when answers are wrong, check what the tool returned first.

42. Where the human fits in the loop · Bullets
Draw the three positions on the loop by hand if you have a whiteboard: before the act, after the final answer, and as an exit from the loop. The lab's last column, 'Human's role', is a first draft of this.

43. 5. Examples from administration and operations · Section · minute 85

44. Five tasks, five designs · Table
Go row by row and ask the room to justify each design using 'who decides the steps' and 'how much judgement'. Row 2 is the course's running example: it is built in n8n on Day 3 and gets an approval gate on Day 5.

45. Notice the pattern in the last column · Callout
This sentence answers the fear in the room about jobs. The work that goes is the copying and chasing nobody enjoys; the deciding stays, and becomes more visible. It is also the ethical frame for Day 6.

46. Preview: when one agent is not enough (Day 5) · Figure · figure: multi-agent-system
Optional if time is short. Plant the vocabulary: supervisor, worker, critic. Day 5 covers sequential, parallel and supervisor patterns and when multi-agent is overkill (usually).

47. Preview: a multi-agent architecture · Figure · figure: multi-agent-architecture
One sentence: more agents does not mean a new kind of thing, it means the same four parts repeated with hand-offs between them. Cut this slide if running behind.

48. Practical: Meet a real agent · Section · minute 90
Start the gateway if it is not running (hermes gateway status). Share the worksheet PDF in the chat. This is the moment the four parts stop being a diagram; keep it light and let participants play.

49. An always-on personal agent · Figure · figure: memory-types
One minute. Name the four parts as you point at the bullets. Say that OpenClaw is the other well-known one and that both are demonstrated properly on Day 5; today you only meet one.

50. Join the agent · Join (QR)
Three minutes. Walk the room. Anyone without Telegram can pair with a neighbour. Replace the handle with your real bot username before the session.

51. Four tasks (10 minutes) · Steps
Let them work. Watch the gateway terminal for the search tool calls and read one aloud. Note who gets a correct recall in Task 1 and whether anyone breaks the guard in Task 4; you will use both in the debrief.

52. What you saw · Two columns
Five minutes. Ask for one observation per task and map it to the right-hand column. A failed guard in Task 4 is the best outcome of the day: it shows why controls and security get two full sessions.

53. Why this is a demo agent, not your department's agent · Callout
Close the practical by placing it in the course. This also keeps faith with Module 9, which says deploying an always-on agent is an IT and security project, not a lab.

54. Practical lab: your task inventory · Section · minute 108

55. Lab steps · Steps
Share the handout and the spreadsheet template in the chat. Walk through the five steps in eight minutes; the inventory itself is homework. The common mistake is tasks that are too vague. Step 4 is where participants get their first taste of using an assistant as a colleague.

56. Step 4: the challenge prompt · Prompt text
Read it aloud and point at the parts: a role, a description of the input, a task, an output order, a constraint ('be brief'). Say: 'Tomorrow in Session 2 you will learn why this prompt is shaped like this.'

57. Deliverable · Callout
Be explicit that this document is reused on Days 2, 3 and 5. Participants who skip it will have nothing to map tomorrow.

58. Knowledge check · Section · minute 116

59–60. Question 1 · Quiz
Q: Which statement best describes an agent, as opposed to an automation?
Answer: B. In an automation every step is fixed in advance; in an agent the model chooses which tool to use next and when to stop.

61–62. Question 2 · Quiz
Q: Which of these is NOT one of the four building blocks of an agent?
Answer: D. The four parts are model, tools, memory and instructions. A dashboard is a way to monitor, not a part of the agent.

63–64. Question 3 · Quiz
Q: Sorting support emails into billing, technical and sales is best built as a...
Answer: B. The path is fixed (read, classify, route) and only one step needs a model. That is the workflow-with-AI-step pattern.

65–66. Question 4 · Quiz
Q: Why does an agent need an iteration limit?
Answer: B. Without a limit a stuck agent burns money and may take more actions than intended. The limit is a basic control.

67–68. Question 5 · Quiz
Q: Which task should generally stay with a person rather than be automated?
Answer: C. Automation removes copying and chasing. Decisions involving money, people or legal exposure need human accountability.

69. Key takeaways · Bullets
Read the four takeaways. Then say what the next session, Session 2, is: writing instructions (prompts) that produce the same shape of answer every time, and building a first assistant that knows your documents.

70. Further reading · Figure · figure: local-book
Optional reading only. The online module repeats everything in this deck, with the knowledge check scored in the browser, so participants can revise alone.

71. Latest reading on agents at work (2026) · Latest articles
Optional two minutes, or leave for self-study. Each item links a headline from the last few months to something taught today. Instructors may swap in newer pieces; keep the pattern: source, date, one-line takeaway tied to the session.

72. References · References
Leave on screen during questions. All figures in this deck are from Alammar and Grootendorst, Hands-On Large Language Models (O'Reilly, 2024), credited on each slide; the same figures appear in Grootendorst's online Visual Guide.

Practical setup summary (Hermes Agent on Telegram)

- Install Hermes Agent on the course droplet (hermes-day1, Ubuntu 24.04, 2 GB, kept for the whole course) with the one-line installer; choose a free provider (Google AI Studio Gemini Flash) with hermes config set model google/gemini-2.5-flash.
- Persona in ~/.hermes/SOUL.md: classroom agent, remembers first name and department only, says it cannot act when asked to send or pay, declines to ignore its instructions.
- Tools: web search only (hermes tools); terminal, files, code and image toolsets disabled.
- Telegram: create the bot with @BotFather, hermes gateway setup, hermes gateway to start; test the four worksheet tasks from your own phone.
- Access: open bot while the gateway runs (started for the practical, stopped right after); or group mode with TELEGRAM_GROUP_ALLOWED_CHATS if open bots are not allowed.
- After each session: hermes gateway stop; delete ~/.hermes/memories/USER.md and ~/.hermes/sessions. Server and bot stay for the whole course and are removed after Day 8.
- Fallback: two-minute screen recording of the four tasks.

Full step-by-step instructions, persona text and fallbacks: **Day1_S1_Hermes_Setup_Guide.pdf** (in this package).

Troubleshooting and cuts

If	Then

Participants have no assistant account	Pair them with a neighbour for the lab; the deliverable is a document, not a chat log. Remind them of the pre-course tool list.
Assistant refuses or gives a very long answer in the demo	Add 'Answer in under 60 words' and repeat. Use it to foreshadow constraints in Session 2.
Participants say 'everything in my job needs judgement'	Ask them to split one task into its steps (read, decide, copy, send, chase). Usually only the 'decide' step needs judgement; the rest is automation.
Wi-Fi, Telegram or the model provider fails during the practical	Play the two-minute fallback recording of the four tasks and run the debrief from it. Keep the timing.
Someone breaks the guard in Task 4 and the agent reveals another participant's department	Say so openly. Only first names and departments are in memory by design. Use it as the first prompt-injection example and point to Day 6.
Running behind at the break	Cut the two multi-agent preview slides and the Toolformer slide, and compress the examples table to three rows. Do not cut the practical or the lab kickoff.

Latest articles used on the slides

- **How AI agents will reshape digital workplace IT operations**. CIO Dive, by Autumn Stanish (Gartner), 19 March 2026. ciodive.com/news/agentic-AI-IT-operations-gartner-digital-workplace/814244
  Gartner expects nearly a third of organisations to run 80% of digital-workplace services autonomously by 2030, from zero in 2025; 84% of IT leaders say extra controls will be needed to govern agents. Controls are the job, not the afterthought.
- **State of AI trust in 2026: shifting to the agentic era**. McKinsey, 25 March 2026. mckinsey.com/capabilities/tech-and-ai/our-insights/tech-forward/state-of-ai-trust-in-2026-shifting-to-the-agentic-era
  Adoption is moving from pilots to scaled, increasingly agentic deployments, but only about 30% of organisations reach a mature level of strategy, governance and agent controls. The gap this course fills.
- **Future of Work with AI Agents: auditing automation and augmentation potential across the U.S. workforce**. Stanford (Shao, Brynjolfsson, Yang et al.), arXiv 2506.06576, revised 1 February 2026. arxiv.org/abs/2506.06576
  An audit of 844 tasks in 104 occupations: workers want agents on the information-heavy, repetitive tasks and want to keep the interpersonal and judgement ones. Exactly the automation-versus-keep-human split in today's lab.
- **Context Engineering: from prompts to corporate multi-agent architecture**. V. Vishnyakova, arXiv 2603.09619, 10 March 2026. arxiv.org/abs/2603.09619
  Argues that managing what an agent knows (its context) is a discipline of its own, with a four-level maturity model. The 'memory' and 'instructions' parts of today's four-part agent, taken seriously at company scale.

References

1. Hermes Agent documentation and source (Nous Research, MIT licence). hermes-agent.nousresearch.com/docs · github.com/NousResearch/hermes-agent

2. Al-Banna, A. (2026). AI Administrator, Module 1: Agents and Automation, Explained. albanna-tutorials.com/ai-administrator/01_agents_and_automation.html
3. Al-Banna, A. (2025). LLM Agents, explained visually (tutorial). albanna-tutorials.com/llm_agents.html
4. Grootendorst, M. (2025). A Visual Guide to LLM Agents. newsletter.maartengrootendorst.com (online version of the same diagrams)
5. Alammar, J. and Grootendorst, M. (2024). Hands-On Large Language Models. O'Reilly Media
6. Yao, S. et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023
7. Schick, T. et al. (2023). Toolformer: Language Models Can Teach Themselves to Use Tools. NeurIPS 2023
8. Anthropic (2024). Introducing the Model Context Protocol. modelcontextprotocol.io
9. Anthropic (2024). Building Effective Agents. anthropic.com/research/building-effective-agents
10. IEEE CS Region 8 AI Caravan (2026). AI Administrator Track Curriculum. aicaravan.org

Figure: Alammar, J. and Grootendorst, M. (2024). Hands-On Large Language Models. O'Reilly Media. Figures also appear in the course tutorial albanna-tutorials.com/llm_agents.html.
